Historical Response Selection for Sampled FX Net-Position Peaks

A fixed comparison and an exploratory state-retrieval extension

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Abstract

Historical peak forecasts depend on which episodes are retained and how their responses are transferred to the current state. We compare empirical quantile constructions for the maximum absolute source-reported USD net position over the current and next H records, using an aggregate client-flow proxy from one foreign-exchange market maker. An original fixed comparison is extended by signed-state nearest-neighbor retrieval after the evaluation interval had been inspected. The primary comparisons use 6,625 common issue origins, a .95 quantile preference and fully completed labels; the extension selects 360 or 180 episodes from a rolling pool of 720. At H=6, historical headrooms have 2.08% lower mean pinball loss than raw current-anchor scenarios, with more breaches and larger overruns; the ranking reverses at H=24. In the extension's fixed primary comparison, nearest360 retrieval reduces floored native-peak-quantile loss by 31.83% relative to recent360 retrieval, but remains worse than the original full-window headroom and scenario rules. All 12 matched retrieval loss contrasts are negative, while every matched scenario contrast has more breaches and greater mean positive overrun. Elementary common-set bounds explain how nearby starting positions can make representations agree without certifying accuracy. The findings identify finite-record forecast tradeoffs, with exploratory retrieval evidence and no population superiority inference. The recorded proxy and passive forecasts do not establish conditional coverage, independent external validity or policy, profit and funding effects.

1. Introduction

A forecast of a future position peak depends on how past episodes are made comparable with the present. One can retain each episode's original peak, retain its increase above its own starting magnitude, or translate its signed increments to the current position before calculating a peak. A further choice is which episodes to retain. Pooling recent episodes and retrieving episodes with nearby starting positions need not produce the same tail forecast, even when both use an unweighted empirical quantile.

We study these choices for a recorded scalar USD net position formed from a sample of a foreign-exchange market maker's client flow. The outcome is the largest absolute reported position over the current and next H records. This is a measure of the scale of the recorded net-position proxy. It cannot reconstruct the dealer's gross currency exposures, hedged book, settlement cash or funding requirement. Our issued levels are passive forecasts under the policy that generated the observations; their scores do not estimate the profit or capital effect of changing quotes, hedges or risk limits. Inventory-sensitive quoting affects future fills and inventory in stated dealer models (Avellaneda and Stoikov, 2008); the policies behind this recorded proxy are not observed here.

The study has two distinct design histories. The original, fixed comparison evaluated historical headrooms, reanchored increment envelopes and three simple references on 6,625 common issue origins. The present extension was formulated after those results had been inspected. It crosses three response representations with five fixed historical selection sets, comparing nearby signed starting positions with an equal number of recent episodes. We published the extension's complete menu and primary contrast before calculating the added forecasts. Every new comparison nevertheless remains exploratory on the already inspected interval.

The extension addresses a specific weakness of the original comparison: applying every representation to a single rolling pool does not show whether a result survives a simple, mature form of state-based historical retrieval. Its primary comparison uses the floored empirical quantile of native peaks, with the 360 nearest starting signed positions versus the 360 most recent origins, both drawn from the same 720 fully completed candidate episodes. The response horizon is six subsequent records. A second, fixed count and a 24-record horizon provide sensitivity evidence. They cannot replace an unfavorable primary result.

The contribution is a bounded application comparison with an explicit construction mechanism. For a common selected set and rank, elementary bounds relate differences between the three issued levels to the historical

starting-position distances. Exact signed-anchor matches make all three constructions agree in real arithmetic. This explains why state matching can reduce a representation difference; it does not establish which forecast is accurate. The empirical analysis therefore reports loss together with capacity, breaches, overrun, unused allowance, selected age and anchor distances. The question is whether a transparent finite retrieval rule changes the observed tradeoff, rather than whether geometric agreement certifies a financial tail model.

The original primary comparison favors headrooms by 2.08% of scenario loss, with more breaches and larger overruns. The extension's primary nearest-state comparison reduces native-peak loss by 31.83% relative to an equal-count recent reference, but does not beat the original full-window leaders. All matched scenario-retrieval score gains coexist with more breaches and greater mean positive overrun. These distinctions are central to the interpretation: a useful retrieval comparison is not a new best-performing risk model or a policy benefit.

2. Closest work and mature method transfer

The closest empirical lineage is Wisniewski, Lindsay and Lindsay (2020), who use this public source for next-hour signed-position interval forecasts. Their methods combine rolling calibration with several regression engines, lagged positions and calendar or trading-session features. Chapter 6 of Wisniewski's doctoral thesis continues endpoint prediction on the same source and evaluates scale-normalized and regularized constructions. These works establish prior financial forecasting on these observations. The response here is instead a multi-record absolute maximum, including the issue record; their published interval scores consequently cannot be used as numerical benchmarks for this target. Our one-state retrieval rule also does not reproduce or displace their richer information sets.

Financial path extrema have a separate established literature. MaxVaR, by Boudoukh, Richardson, Stanton and Whitelaw (2004), studies the worst mark-to-market loss relative to initial wealth over a monitoring horizon. Van den Hoogen (2017) studies maximum cumulative payment-cash deficits, while Liao and Theodosopoulos (2005) consider maximum collateral-adjusted contingent exposure and marking decisions. Their states, units and financial uses differ from a two-sided absolute net-position maximum. They make a broad claim of a first financial peak forecast untenable, while leaving a legitimate, narrower application question about the particular recorded response.

The method components are mature. Hull and White (1998) update historical shocks using predicted volatility; Barone-Adesi, Giannopoulos and Vosper (1999) propagate standardized historical innovations through stated conditional dynamics and portfolio revaluation. The scenario representation here performs only raw translation of observed net-position increments. It fits neither a volatility filter nor a dynamics model. Biau and Patra (2011) provide a directly relevant nearest-neighbor empirical-response-quantile construction within a stationary-ergodic sequential prediction strategy. Our fixed rolling window, one current signed level and delayed, overlapping peak labels do not implement that complete strategy or satisfy its assumptions merely by using neighbors. We transfer the finite retrieval operation and assess it on a common response; no consistency or coverage theorem is inherited.

The resulting difference from the nearest papers is thus substantive but limited: a different financial response, explicit completed-label timing, a fixed crossing of historical selection and response representation, and an elementary explanation of when those representations approach one another. It is not a new quantile estimator, a new filtered historical simulation method, an independent market-maker validation or a demonstrated improvement over the strongest conditional risk models.

3. Data, recorded response and completed information

The input is the one-hour member of version 2 of the Net Position Time Series Dataset, with 18,576 records from 5 February 2014 01:00 through 24 January 2017 00:00. Wisniewski et al. (2020) describe sampled aggregate client-flow positions across five liquid FX pairs. Only datetime and NetPosUsd are used. A netted USD scalar does not reconstruct currency-level gross positions, hedge transactions, collateral or cash movements. The source's timezone and historical publication/vintage convention are unverified.

There are 18,420 one-hour and 155 forty-nine-hour adjacent gaps. We retain the file's chronology without sorting, deleting paths across gaps or interpolating missing intervals. Consequently H counts subsequent records, rather than elapsed hours. Let N_t be the signed source-reported position, observed before issue. Define

$$B_t=|N_t|;\quad P_{t,H}=\max_{0\le h\le H}|N_{t+h}|;\quad G_{t,H}=P_{t,H}-B_t.$$

Including h=0 gives the known floor B_t and nonnegative headroom G. The peak includes an interior recorded spike even if the final position returns to its starting value. It does not bound an unobserved excursion between records. The retrospective issue convention treats N_t as available immediately before issue, rather than establishing historical live-publication availability.

At issue t the extension's eligible pool is $I_t=\{t-H-719,...,t-H\}$. Each of its 720 historical origins i has a response ending at $i+H\le t$. The latest completed label may therefore include the already observed issue record. Earlier evaluation outcomes enter later pools only after completion. Adjacent peaks and rolling pools overlap; completed-label eligibility does not make them independent.

For each eligible origin retain its native peak $P_{i,H}$, headroom $G_{i,H}$ and directional increment extrema

$$a_{i,H}=\min_{0\le h\le H}(N_{i+h}-N_i);\quad b_{i,H}=\max_{0\le h\le H}(N_{i+h}-N_i);\quad a_{i,H}\le 0\le b_{i,H}.$$

These extrema determine the absolute peak after raw translation of the sampled increment path. They are not merely its two endpoints, and their deterministic sufficiency does not specify the next path's conditional law.

4. Historical selection and response representation

The extension uses five fixed sets at each issue. Recent sets take the last m eligible origins. Nearest sets take the m smallest signed-state distances $|N_i-N_t|$, preferring the most recent origin when distances tie. Historical response values and future query outcomes never resolve distance ties. Selected paired rows retain chronological order.

Table 1. Fixed historical sets drawn from the same 720 completed candidate episodes. The rank is k=ceil((m+1)19/20).

Set	Membership	m	k
full720	All eligible	720	685
recent360	Most recent	360	343
nearest360	Nearest signed state	360	343
recent180	Most recent	180	172
nearest180	Nearest signed state	180	172

For a selected set J of m rows, Q_k denotes the kth unweighted order statistic with duplicates retained. There is no interpolation or jitter; all configured corrected ranks are available. At these particular counts and $\tau=.95$ the selected order statistic is the upper endpoint of the empirical pinball-minimizer interval (Appendix C). This numerical rank convention establishes no coverage theorem for the dependent historical pool.

The three response representations issue the following levels, with H suppressed on historical labels:

$$F_J=B_t+Q_k\{P_i-|N_i|: i \text{ in } J\}.$$

$$S_J=B_t+Q_k\{\max(N_t+b_i,-N_t-a_i)-B_t: i \text{ in } J\}.$$

$$K_J=\max\{B_t,Q_k\{P_i: i \text{ in } J\}\}.$$

F retains headroom relative to historical starting magnitude; S translates signed increments to the current position; K retains native peaks subject to the current floor. K is the method labeled L in the original artifacts. The files call an issued forecast level C its 'capacity'; throughout the paper this is a passive reported level. All three already use current information: F/K use B_t, and S also uses signed N_t. Retrieval changes historical membership. Equal-count recent and nearest sets may differ in age, scale, regimes and dependence.

The original five-rule comparison also includes the radius reference $R=B_t+Q_k\{\max(b_i,-a_i)\}$ and the known-floor reference $P0=B_t$, on its full n-row pool. The extension implements only F/S/K. Stored float64 S subtracts the floor before ranking and adds it afterward, preserving the original operation order and its rounded decisions.

An elementary construction bound explains what matching can control. For one common nonempty set J and the same valid rank, put $r_{abs}=\max_{i \text{ in } J}||N_i|-B_t|$ and $r_{signed}=\max_{i \text{ in } J}|N_i-N_t|$. In real arithmetic,

$$|F_J-K_J|\le r_{abs};\quad |S_J-K_J|\le r_{signed};\quad |F_J-S_J|\le r_{abs}+r_{signed}\le 2r_{signed}.$$

Appendix A proves the bounds using floor projection and order-statistic stability. If every selected signed anchor equals the query, all three reports agree; if only every selected magnitude equals the query magnitude, F=K while S can differ. Plateaus can also give equal F/S reports at distant anchors. Nearest retrieval minimizes the largest selected signed distance among equally sized subsets, but need not minimize actual report differences or loss. These common-set bounds do not compare forecasts built from different selected histories, and they contain no statement about future calibration.

5. Design histories, score and reporting

The original specification was recorded at 05:46:37 UTC on 21 October 2026 and publicly committed before its target-specific forecasts and losses were calculated. It fixes H=6, n=720, τ=19/20 as primary, with six one-factor changes: H=12 or 24, n=480 or 1440, and τ=.90 or .99. The five rules and seven settings yield 35 original summary rows. This is a retrospective specification exercise with prior source-literature knowledge, not external preregistration or prospective collection.

The extension was conceived after those results had been inspected. Its human-readable protocol and hash-pinned machine specification were publicly recorded in commits 63703b9 and 69d5bb6 before the added forecasts. They fix H=6 and 24, the five sets in Table 1 and F/S/K: 15 configurations per horizon, 30 in total. The six full720 F/S/K rows repeat original F/S/L references. Recording the extension before execution limits further discretion but cannot turn the old interval into an unseen holdout.

Both designs use the same 6,625 issue origins: indices 11927 through 18551 inclusive, from naive source timestamp 1 January 2016 00:00 through 23 January 2017 00:00. The common original full-menu rule retains the future records required by H=24. Earlier evaluation labels become usable later only when completed. Changing H alters both the future peak and the end of the eligible historical archive, so cross-horizon comparisons do not isolate the maximum operator alone.

For stored error e=P−C, pinball loss is

$$\rho_\tau(e)=\tau \max(e,0)+(1-\tau)\max(-e,0); \quad \tau=19/20 \text{ in the extension.}$$

The underprediction weight is nineteen times the overprediction weight. These are stipulated forecast preferences, not estimated funding prices, shortage costs or regulatory coefficients. Quantile-consistent scoring targets the corresponding functional under suitable finite expected-loss conditions (Gneiting, 2011); it does not establish conditional correctness of an empirical historical rule. Original rows with different τ express different objectives and are not pooled into an across-preference winner.

The original primary contrast is mean loss(F) minus mean loss(S) in its primary setting. The extension's separate primary contrast is mean loss(nearest360K) minus mean loss(recent360K), at H=6; a nonnegative mean would fail its finite-record improvement criterion. H=24 repeats retrieval in a context adverse to the original F/S comparison, and m=180 is fixed sensitivity. Neither may replace an unfavorable primary contrast. All 12 nearest-minus-recent and 10 same-set F-minus-S contrasts are reported without a score-sign acceptance gate.

For an issued level C≥B, extra allowance is C−B. Positive overrun and unused allowance give the score decomposition

$$O=(P-C)^+; \quad U=(C-P)^+; \quad \rho_\tau(P-C)=\tau O+(1-\tau)U, \text{ where } z^+=\max(z,0).$$

Every configuration reports mean loss, level and extra allowance; strict breach count/rate; mean O; conditional overrun with its own breach denominator; and mean U. Strict breach is P>C, with no tolerance changing its sign. Conditional overrun averages O only over that rule's breach events; zero breaches make it undefined. The target can have an atom at B, so even a true quantile's strict exceedance need not equal 1−τ. Nominal breach proximity is not a calibration certificate.

Geometry reports both anchor radii, mean selected age in records, the mean absolute F/S gap and exact equal-capacity counts. Primary monthly retrieval contrasts retain all 13 inclusive calendar months at both H; groups below 30 origins retain counts/status while suppressing metrics. These are descriptive groups, not independent replications or an adaptive strategy. No new population test, confidence interval or bootstrap is attached to the extension. Original

block-resampling profiles concern only the original full720 F/S contrast; Appendix B preserves their descriptive status.

6. Original comparison: score advantage with adverse tradeoffs

Table 2. Original primary H=6, n=720, τ=.95 results. Monetary columns are source-reported USD millions; breach is a percentage. K is the original L. Every rule has 6,625 origins.

Rule	Loss	Level	Breaches	Breach %	Overrun	Unused
F	0.441774	22.645674	332	5.011	0.166531	5.671391
S	0.451149	23.829232	254	3.834	0.116728	6.805145
R	0.527544	26.400452	122	1.842	0.064562	9.324199
K (old L)	0.925885	31.731323	489	7.381	0.196359	14.786868
P0	1.813620	15.231741	5102	77.011	1.909073	0.000000

Historical headrooms F have mean loss USD 441,774.04 versus USD 451,148.90 for raw reanchored scenarios S. The paired mean is −USD 9,374.86, or −2.08% of S loss. It is aggregated from per-origin loss differences before display rounding. All original sensitivities and methods are retained in Appendix B and the public aggregates.

The lower score accompanies 332 strict breaches for F versus 254 for S, and greater mean positive overrun (USD 166,531.06 versus USD 116,728.04). Conditional overrun is USD 3.323097 million over 332 events for F and USD 3.044580 million over 254 for S. Lower unused allowance offsets the larger overrun at the stipulated weights. It is a score tradeoff rather than safety dominance or a measured financial saving.

R has fewer breaches and smaller unconditional overrun, but larger mean capacity and score. K has the greatest mean capacity yet 489 breaches, illustrating why an average capacity ordering does not imply pathwise dominance. P0 has zero extra and unused allowance, but 5,102 breaches and the largest loss. Conditional means must be read with these different event sets, rather than ranked as if they describe the same tail observations.

The F/S direction reverses at H=24: F loss is USD 777,068.84 and S loss USD 748,383.54, with paired mean +USD 28,685.30. The τ=.90 difference is also positive but small (+USD 144.23), retained without rounding it to zero or declaring equivalence. Original monthly means favor F eight times and S five times. Low and middle pre-cutoff magnitude groups favor S, while the high group favors F and accounts for 5,445 of 6,625 origins. Those descriptive composition patterns do not identify a causal state effect or constitute a formal stationarity test.

All four original geometric-block resampled-mean ranges cross zero. They are descriptive profiles, not population confidence intervals, superiority probabilities or proof of equality. The original H24 reversal and these limits remain unchanged by a later favorable retrieval contrast, which asks a different question.

7. Exploratory historical selection

The fixed primary retrieval criterion is met on this inspected interval. At H=6, m=360, nearest360K loss is USD 594,811.09 versus USD 872,515.55 for recent360K; the direct paired difference is −USD 277,704.47, a 31.83% reduction relative to recent360K. Both original full720 F and S remain lower. At H=24, nearest360K improves over recent360K from USD 1,246,630.42 to USD 960,696.45, while original full720 S remains lower at USD 748,383.54. The large relative gain therefore improves a weak native-level baseline, without establishing a new best forecast.

Table 3. Complete extension loss menu and same-set F-minus-S paired mean, in source-reported USD millions. Full720 rows repeat original references. Differences are computed before display rounding.

H	Historical set	F loss	S loss	K loss	F−S mean
6	full720	0.441774	0.451149	0.925885	-0.009375
6	recent360	0.454089	0.467678	0.872516	-0.013589
6	nearest360	0.448664	0.448731	0.594811	-0.000067
6	recent180	0.485866	0.478940	0.785878	+0.006926
6	nearest180	0.468133	0.459592	0.525471	+0.008541
24	full720	0.777069	0.748384	1.197833	+0.028685
24	recent360	0.807496	0.824047	1.246630	-0.016551
24	nearest360	0.791192	0.770354	0.960696	+0.020838
24	recent180	0.858214	0.895718	1.284092	-0.037504
24	nearest180	0.847734	0.827305	0.963931	+0.020429

All 12 fixed nearest-minus-recent score contrasts are negative (Figure 2). Absolute reductions are largest for K in each horizon/count pair: about USD 260,407–320,161, compared with USD 18,947–68,413 for S and USD 5,425–17,733 for F. These are correlated comparisons sharing outcomes and historical pools, not 12 independent confirmations. Nearest360S at H=6 slightly improves its full720S counterpart. Nevertheless, no added configuration beats the better original full720 F/S rule at the corresponding H: F at H=6 and S at H=24. This observed menu comparison is not a new prespecified model-selection criterion.

For the primary K pair, mean capacity falls from USD 29.854484 million to USD 25.224925 million; mean unused allowance falls from USD 12.950501 million to USD 8.274716 million. Breaches fall from 637 to 507 (9.615% to 7.653%), and mean positive overrun falls from USD 236,832.10 to USD 190,605.57. Conditional overrun nevertheless rises from USD 2.463128 million over 637 events to USD 2.490655 million over 507. The denominators and breach sets differ. None of these quantities is an observed funding payment or conditional-calibration result.

Every matched S retrieval comparison has the opposite breach/overrun tradeoff: lower loss but more strict breaches and greater mean positive overrun. At H=24, m=360, nearest S loss falls from USD 824,046.89 to USD 770,353.74, while breaches rise from 320 to 430 and mean positive overrun from USD 190,842.79 to USD 283,299.56. Lower capacity and unused allowance offset the overrun increase under the score. The same adverse direction holds for both counts at both horizons. Describing all 12 score gains as uniform risk reduction would be inaccurate.

For F, nearest retrieval reduces breach counts and mean positive overrun in all four matched comparisons, but conditional mean overrun rises in all four. At H=24, m=180, it rises from USD 4.650289 million over 588 breaches to USD 5.349645 million over 454. These conditional means summarize different breach sets; they do not identify an increase in severity on the same events.

State matching also changes the construction geometry. At H=6, m=360, mean signed radius falls from USD 30.399817 million for recent retrieval to USD 12.365836 million for nearest retrieval, mean absolute F/S gap from USD 1.256683 million to USD 0.477250 million, and exact equal-capacity count rises from 1,804 to 4,297. The paired F−S loss mean becomes −USD 66.78, rather than recent360's −USD 13,589.40. Its small magnitude is an observed difference, not statistical equivalence.

At H=24, m=360, the smaller nearest F/S report gap coexists with a loss-order reversal: recent360 favors F by USD 16,550.54, whereas nearest360 favors S by USD 20,837.78. Reducing nearest count from 360 to 180 makes radii and F/S gaps smaller at both H but makes both F and S losses larger. Thus geometric agreement cannot select a predictive optimum. The complete geometry and age table is retained in Appendix B.

Nearest rows are older on average than equal-count recent rows. At m=360, mean age is 328.35 versus 185.50 records for H=6, and 352.09 versus 203.50 for H=24; at m=180 the nearest ages are 301.27 and 331.82 versus 95.50 and

113.50. State proximity, age and potentially regime/scale/dependence composition change together. The comparison does not identify a pure causal contribution of state information.

The primary monthly K contrast favors nearest retrieval in 12 of 13 months at each H (Figure 3). January 2016 is adverse at both: +USD 3,379.20 for H=6 and +USD 520,447.46 for H=24. All month counts exceed the fixed display threshold. Origin-count-weighted monthly means reconcile to the global paired means; months remain descriptive groups within the same record, and the adverse month is not excluded.

8. Interpretation and limits

The added comparison supports a narrow application result: a mature one-dimensional retrieval operation improves these equal-count recent historical rules on this interval, particularly the native-level baseline in absolute score units. It also reduces representation differences, consistent with finite geometry. It does not displace the original full-window leaders or establish that one signed position is a sufficient forecasting state. Wisniewski et al. (2020) and the thesis already use richer lag/calendar/scale information; the present extension does not benchmark the strongest conditional regressions or fitted filtered historical simulation.

Raw historical translation and conditional path generation are different operations. Appendix A gives a known-linear-dynamics illustration: innovation replay adjusts an anchor difference by a horizon-dependent factor, whereas raw translation keeps the whole difference at every step. The identity is deterministic and no coefficient is fitted here. It explains why exact revaluation of a historical increment envelope is weaker than having its correct future conditional law; it does not establish the source's inventory-control mechanism.

The score preference is part of the decision problem. At the declared 19:1 ratio, less unused allowance can compensate for greater overrun. A different loss or origin weighting would ask a different question, rather than repair missing financial-cost observations. Conditional severity compares method-specific breach sets, and sampled net position remains a proxy. Strong forecast performance on that proxy would still not measure cash adequacy, gross exposure, P&L or the causal effect of a changed policy.

External validity is the principal empirical limitation. One aggregate 2014–2017 source cannot establish robustness across dealers, subsequent regimes or interventions. Reexecutions, a fresh identical-hash download and a same-source thesis are reproducibility evidence rather than independent records. A legitimate additional inventory record would need its own aggregation, units, sampling and release-time audit. Price returns and weekly position reports are not interchangeable inventory measurements.

Overlapping future maxima, shared historical episodes and persistent positions create dependence. The stationarity, mixing, conditional-law and tail-information conditions needed for broader inference have not been established. Fixed neighbor counts are not counts of independent tail observations, and more favorable cells do not supply independent confirmations. Neither an elementary construction bound, a corrected empirical rank nor an observed breach frequency yields coverage. The extension remains exploratory despite its before-execution freeze.

9. Conclusion

Historical row selection and response representation lead to distinct, observable forecast tradeoffs for sampled client-flow net-position peaks. Original headroom pooling improves the primary score over raw scenario reanchoring by 2.08%, with greater breaches/overrun and an adverse H24 reversal. Exploratory signed-state retrieval improves equal-count recent rules, including a 31.83% primary reduction for the native-peak baseline, while leaving the original full-window leaders ahead and increasing scenario breaches/overrun in every matched case. Finite geometry explains when constructions approach one another without deciding which predicts better. The evidence provides a reproducible, limited application comparison; stronger scientific claims require genuinely independent validation and more informative financial/conditional benchmarks.

Appendix A. Finite geometry and its predictive limits

A.1. Common-set projection bounds

These are elementary finite-array arguments, rather than new asymptotic or coverage results. Fix a nonempty common selected set J, an available rank $1\le k\le|J|$, query $x=N_t$, floor $B=|x|$ and completed historical windows. All identities in this appendix concern real arithmetic. Write $B_i=|N_i|$, $d_{i,h}=N_{i+h}-N_i$, with $d_{i,0}=0$, and define

$$A_i=B+P_i-B_i;\quad T_i=\max_{0\le h\le H}|x+d_{i,h}|;\quad C_B(z)=\max(B,z).$$

Both A_i and T_i are at least B. The projection C_B is nondecreasing and 1-Lipschitz. It commutes with a finite order statistic, as does a common translation. Hence $F_J=Q_k(A)$, $S_J=Q_k(T)$ and $K_J=Q_k(C_B(P))$. Since $C_B(A_i)=A_i$,

$$|A_i-C_B(P_i)|\le|A_i-P_i|=|B-B_i|.$$

The reverse triangle inequality bounds each path value's change under a starting-position translation by $|x-N_i|$. Taking a maximum gives $|T_i-P_i|\le|x-N_i|$. Applying the same floor projection preserves that bound because $T_i\ge B$. Finally, paired order statistics are nonexpansive in the maximum coordinate difference: if $|u_i-v_i|\le r$ for every i, then $u_i\le v_i+r$ and conversely; monotonicity and common translation give $|Q_k(u)-Q_k(v)|\le r$. These observations prove the two K bounds in Section 4. The triangle inequality and $||N_i|-|x||\le|N_i-x|$ prove the stated F/S bound. Keeping the rank fixed while changing one side's historical rows does not by itself justify these common-set comparison bounds.

Among all size-m subsets, selecting the m smallest signed distances minimizes the largest selected distance: any alternative must include an element at least as far away as the mth ordered distance. This finite minimax fact, with the declared tie convention, concerns the signed radius. It does not minimize the absolute-anchor radius, the sum of the two radii, the actual report gap or a forecast loss. For every fixed realized y, pinball loss is $\max(\tau,1-\tau)$-Lipschitz in its report, so a report-gap bound also bounds the absolute loss difference. Such stability supplies no sign for the realized or expected loss difference.

A.2. Envelope plateaus and the role of signed anchors

Put $u_i=b_i\ge0$ and $v_i=-a_i\ge0$. The headroom of an increment envelope at any anchor z is

$$g_i(z)=\max(z+u_i,-z+v_i)-|z|;\ \text{for } z\ge0 \text{ it equals } \max(u_i,v_i-2z),\ \text{and for } z\le0 \text{ it equals } \max(u_i+2z,v_i).$$

The branches agree at zero. Each is 2-Lipschitz on its half-line, so the whole function is 2-Lipschitz; its range lies in $[\min(u_i,v_i),\max(u_i,v_i)]$. Applying these two bounds to $g_i(N_i)$ and $g_i(x)$, and then order-statistic stability, gives

$$|F_J-S_J|\le\max_{i\text{ in }J}\min\{2|x-N_i|,|u_i-v_i|\}.$$

For example, $u_i=v_i$ in every row makes $F_J=S_J$ at arbitrary anchors. Nonnegative query/source anchors together with $u_i\ge v_i$ in every row also suffice; the nonpositive analogue uses $v_i\ge u_i$. These are sufficient conditions, not a characterization: pooled order statistics can coincide even when paired rows differ. The range bound also gives $F_J,S_J\le R_J$, where $R_J=B+Q_k\{\max(u_i,v_i):i\text{ in }J\}$. This is a theoretical same-set ordering; no R retrieval configuration is evaluated in the extension, and the ordering does not make R_J preferable under a given loss.

If every $B_i\le B$, then $A_i\ge C_B(P_i)$ and $F_J\ge K_J$. If every $B_i\ge B$, then $C_B(P_i)=P_i\ge A_i$ and $F_J\le K_J$. Equal absolute anchors force F=K but leave signs relevant to S. A coherent full-pool example uses $c>0$, $H=1$, $n=720$, $\tau=19/20$, $t=721$ and $N_s=(-1)^{s+1}c$ for $0\le s\le721$. The mature pool is $J=\{1,...,720\}$, with 360 anchors of each sign and current $N_t=c$. All native peaks equal c and headrooms equal zero, so F=K=c. Raw reanchoring gives 360 peaks c and 360 peaks 3c; rank 685 gives S=3c. Thus $r_{abs}=0$ while $r_{signed}=2c$ and $|S-K|=2c$. This artificial path respects maturity and an available corrected rank; it is not an empirical claim or a claim about which rows nearest retrieval selects.

A.3. The same observed history allows opposite loss rankings

Choose $d>0$, $H\ge1$, $0<\tau<1$ and n large enough that $k=\text{ceil}((n+1)\tau)\le n$. Set $t=H+n-1$ and $N_s=d(t-s)$ for $0\le s\le t$, using $J=\{0,...,n-1\}$. Every selected response ends by t. Each historical peak is its own positive starting value, so $G_i=0$ and F=0. Each increment path equals $-dh$; reanchoring at $N_t=0$ gives S=Hd. Two continuations of this identical observed history yield:

Table A1. Artificial coherent continuations. Symbols are constructed quantities, not source-record estimates.

Future N(t+h), 1≤h≤H	Actual peak	F loss	S loss
0	0	0	(1−τ)Hd
−dh	Hd	τHd	0

Either report can therefore have strictly smaller realized loss under a compatible future. Even a capacity ordering need not hold universally: the artificial history (0,−4,−1,2), with H=1, n=3, τ=3/4 and issue t=3, has rank 3, F=6 and S=5. These constructions establish logical limits, not the conditional law of the recorded series or a population ranking. A deterministic construction can refute an unconditional mathematical assertion while supplying no empirical validation for either rule.

A.4. Raw translation versus replay under known dynamics

For interpretation only, suppose $N_{s+1}=\mu+\varphi(N_s-\mu)+\varepsilon_{s+1}$, with a common known center and coefficient. Replay a historical window's ordered innovations from query x. Subtracting the two recursions and using induction gives

$$M_{i,h}(x)=N_{i+h}+\varphi^h(x-N_i);\quad W_{i,h}(x)=N_{i+h}+(x-N_i).$$

Taking the maximum absolute path value yields

$$|\max_h|W_{i,h}(x)|-\max_h|M_{i,h}(x)||\le\max_{0\le h\le H}|1-\varphi^h|\,|x-N_i|.$$

For $0\le\varphi\le1$ the coefficient is $1-\varphi^H$; at $\varphi=1$ the paths coincide. The identity needs no innovation law, while treating past innovations as representative of future conditional innovations is a separate assumption. Neither the recursion nor that assumption is fitted or validated here. Except in special cases such as φ=1, evaluating the horizon-dependent correction generally requires ordered path information that increment extrema discard. Thus the extrema are sufficient for exact raw translation, not for arbitrary filtered simulation or correct conditional path generation.

Appendix B. Complete settings and descriptive diagnostics

The following tables retain favorable, adverse and small differences. Monetary entries are source-reported USD millions, rounded only for display. Paired means come directly from their accepted aggregate columns; the tables do not recompute forecasts, select settings or create new tests. K denotes original L. Original settings change one factor from H=6, n=720, τ=.95: h12/h24 change H, n480/n1440 change n, and tau90/tau99 change τ. All use the same 6,625 issue origins. Different τ values express different score preferences.

Table B1. Complete original score grid. Δ is the stored paired mean F minus S; negative favors F within that setting.

Setting	F	S	R	K	P0	Δ F−S
main	0.441774	0.451149	0.527544	0.925885	1.813620	-0.009375
h12	0.592518	0.604781	0.689604	1.058614	2.950594	-0.012263
h24	0.777069	0.748384	0.803106	1.197833	4.630827	+0.028685
n480	0.449843	0.460816	0.532515	0.885243	1.813620	-0.010973
n1440	0.443793	0.448275	0.526979	0.931562	1.813620	-0.004482
tau90	0.659643	0.659499	0.763481	1.477201	1.718166	+0.000144
tau99	0.154991	0.158998	0.175486	0.288834	1.889983	-0.004007

The original groups were fixed using issue-time information except the explicitly retrospective future-gap panel. Magnitude cutoffs use absolute positions strictly before the evaluation interval; neither cutoff is reestimated within an evaluation group. Sign and magnitude panels have unequal membership. The index-modulo-six panel is a mechanical phase diagnostic rather than a six-hour sampling experiment. A group below 30 origins retains its count and suppression status; an empty current-zero group does not have a zero mean loss.

Table B2. All original noncalendar paired diagnostic groups. Δ is F minus S at the original primary setting. 'Not displayed' denotes insufficient membership, not an estimated zero.

Panel	Group	Origins	Δ F−S
Sign	negative	5592	-0.010808
Sign	zero	0	Not displayed
Sign	positive	1033	-0.001618
Magnitude	low	420	+0.023570
Magnitude	middle	760	+0.016090
Magnitude	high	5445	-0.015470
Index mod 6	0	1104	-0.012793
Index mod 6	1	1104	-0.016473
Index mod 6	2	1104	-0.005393
Index mod 6	3	1104	-0.005242
Index mod 6	4	1104	-0.005957
Index mod 6	5	1105	-0.010390
Future gaps	gap_free	6289	-0.006642
Future gaps	has_gap	336	-0.060535

Gap-crossing groups classify the realized future path and cannot be used as an issue-time filter. The 336 gap-crossing origins contain only one F breach and one S breach; their conditional severities rest on one event each. Calendar panels use naive source timestamps. Table B3 retains the original F/S month contrast and breach counts alongside the extension's two primary K month contrasts; they are separate contrasts, not interchangeable strategies.

Table B3. All months. Old Δ is original H6 F minus S; new Δ is nearest360K minus recent360K. Breach counts belong to old F and S. Every month meets the display threshold.

Month	Origins	Old Δ	F breaches	S breaches	New Δ H6	New Δ H24
2016-01	504	-0.017183	29	18	+0.003379	+0.520447
2016-02	504	-0.015726	21	18	-0.366978	-0.406803
2016-03	552	+0.040380	42	31	-0.211499	-0.086232
2016-04	502	+0.020270	37	24	-0.522837	-0.589971
2016-05	530	-0.031114	10	7	-0.432595	-0.352734
2016-06	528	-0.035273	16	7	-0.236362	-0.105785
2016-07	504	+0.006976	34	27	-0.213175	-0.494943
2016-08	552	-0.047937	14	12	-0.494498	-0.685697
2016-09	526	+0.002870	31	29	-0.236850	-0.338212
2016-10	506	-0.007725	39	36	-0.230552	-0.135516
2016-11	528	+0.007843	25	24	-0.220045	-0.310768
2016-12	526	-0.032346	15	11	-0.323571	-0.422938
2017-01	363	-0.012823	19	10	-0.041940	-0.286619

Table B4. All extension geometry rows. Mean per-origin anchor radii and mean absolute report gaps are in USD millions; age is mean t−i in records, not hours; equality counts are exact stored F=S comparisons.

H	Set	Abs radius	Signed radius	|F−S|	Equal	Age
6	full720	28.439284	36.604661	1.212343	809	365.50
6	recent360	24.262721	30.399817	1.256683	1804	185.50
6	nearest360	11.346111	12.365836	0.477250	4297	328.35
6	recent180	20.063962	24.372598	1.249574	2914	95.50
6	nearest180	6.506279	7.065549	0.313294	5100	301.27
24	full720	28.603627	36.864344	1.836659	1234	383.50
24	recent360	24.420571	30.694525	2.824054	2412	203.50
24	nearest360	11.566355	12.627357	0.703643	4350	352.09
24	recent180	20.576841	25.146348	3.169128	3342	113.50
24	nearest180	6.782036	7.387771	0.680282	5062	331.82

All ten geometry rows pass the per-origin declared tolerance checks. Nine aggregate maximum excesses are zero; H24/nearest180 records a maximum excess of 1.862645149230957e−9 USD, reflecting floating-point arithmetic rather than an exact-real violation. Its maximum recorded arithmetic tolerance is 1.041112061415106e−6 USD. Comparing these two aggregate maxima alone is not the per-origin verification; the issuer checks every origin and the accepted run records that check. Tolerances do not change breach flags, exact-capacity equality or loss signs.

The original descriptive resampling operates on the single already realized primary F-minus-S loss-difference sequence. For each length ℓ in {120,240,720,1440}, it uses 999 circular resamples: start uniformly at an origin, draw a positive geometric block length with mean ℓ, continue circularly, and truncate only the final block to length M=6,625. Each length uses NumPy PCG64 with SeedSequence([20261020,ℓ]). The table reports ddof=1 resampled standard deviation and noninterpolated order statistics 25 and 975. Constant input is preserved exactly; no recentering or model refitting is performed.

Table B5. All four original resampled-mean profiles, in USD millions where monetary. Expected block counts include the final truncated block. The displayed ranges are not population confidence intervals.

ℓ	ℓ/M	E blocks	Rep. mean	Rep. SD	Order 25	Order 975
120	0.0181	56.2	-0.009774	0.009310	-0.028186	+0.008559
240	0.0362	28.6	-0.009835	0.008948	-0.028279	+0.008127
720	0.1087	10.2	-0.009471	0.007557	-0.025106	+0.004780
1440	0.2174	5.6	-0.009475	0.006679	-0.023037	+0.003596

Politis and Romano (1994) supply the stationary-bootstrap lineage, whose formal results impose conditions on the stochastic process and block-length asymptotics. Here no relevant stationarity/mixing conditions have been established, and large fixed ℓ/M leaves few blocks. The profiles therefore describe a fixed resampling mechanism's sensitivity on this record. They are not applied to any added nearest-state contrast, and their endpoints are neither null-test cutoffs nor superiority probabilities.

Appendix C. Rank, arithmetic and replication conventions

For m sorted responses $z_{(1)}\le\ldots\le z_{(m)}$, the corrected rank is k=ceil((m+1)τ). If k>m, this implementation stops rather than inserting an infinite report or changing the rank. All declared cells have an available rank. The original seven ranks are 685,685,685,457,1369,649,714 in the Table B1 order. In the extension τ=19/20 and m in {720,360,180}, so mτ in {684,342,171} is an integer. The empirical pinball-minimizer interval is $[z_{(m\tau)},z_{(m\tau+1)}]$, including ties; the configured corrected rank chooses its upper endpoint. This particular relationship is not asserted for the original τ=.99 cell, nor does any empirical-minimizer identity establish future coverage.

All forecasts use stored float64 values, duplicate-preserving order statistics and the original operation order. For S, the translated peak is computed first, B is subtracted before ranking, and B is added afterward. Replacing this with a mathematically equivalent differently rounded expression can alter stored capacity ties, strict flags and scores. The scientific tables use direct error P−C; the paired summary uses compensated aggregation of per-origin loss differences, rather than subtracting displayed or independently rounded global means. Empty CSV values and JSON nulls have documented undefined reasons. Finite numerical reconciliation tolerances are kept separate from event definitions.

The extension's per-origin geometry tolerance is 64ε times the maximum of one, current floor, historical anchor magnitudes, native peaks, all three reports and the two anchor radii. The input source, specification and code identities are checked before accepting a run and rechecked after computation. Membership is computed from signed source positions at the completed historical origins; absolute magnitude alone cannot authenticate their signs. Selected age and radii are descriptive properties of that membership, not extra tuned predictors.

The input is version 2's one-hour member, 1hourNetPositionsKaggle.csv, with 18,576 records and 1,286,391 bytes. Its accepted SHA-256 is b8e92c54a76b49990b028252d6a89195e92193990e72f10228f2280a11a967fa. The public Net Position Time Series Dataset is identified by the Kaggle owner/slug davidlindsay1979/net-position-time-series-dataset. The source fingerprint, member/version, acquisition checks and result/code manifests allow identity verification. Public availability is not taken as permission to redistribute the raw observations; this repository supplies acquisition instructions and analysis code.

The original accepted computation took 8.02145 seconds, with peak sampled process RSS 80,936,960 bytes and one observed OS thread. The extension's sole accepted calculation took 10.307238 seconds on its internal process clock, with peak sampled RSS 64,602,112 bytes and one observed thread over 675 snapshots. A process-completion measurement reported 10.35 seconds and maximum RSS 64,634,880 bytes; its scope and precision differ from the internal measurement. These are execution observations under the single-process CPU budget, not general runtime benchmarks. The extension reused the exact accepted local input and did not download or redistribute observations.

The original results remain in results/summary.csv, paired-comparison.csv, stability.csv and diagnostics.csv with their original manifest. The separately frozen extension has summary.csv (30 rows), geometry.csv (10), comparisons.csv (22), monthly.csv (26) and manifest.json under results/revision-20261021/. Six shared reference rows reproduce the original F/S/L numeric columns exactly, with K mapped to L. Reproducing these values on the same hash-identified source verifies execution, not external validation. The root README documents separate acquisition, scientific execution, aggregate-only figure construction and manuscript rebuilding. Document rendering reads approved text and figure assets rather than running financial experiments.

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davidlindsay1979. *Net Position Time Series Dataset*, version 2. Kaggle. Dataset record. Source identity verified 21 October 2026; raw redistribution permission remains unverified.

Figure captions

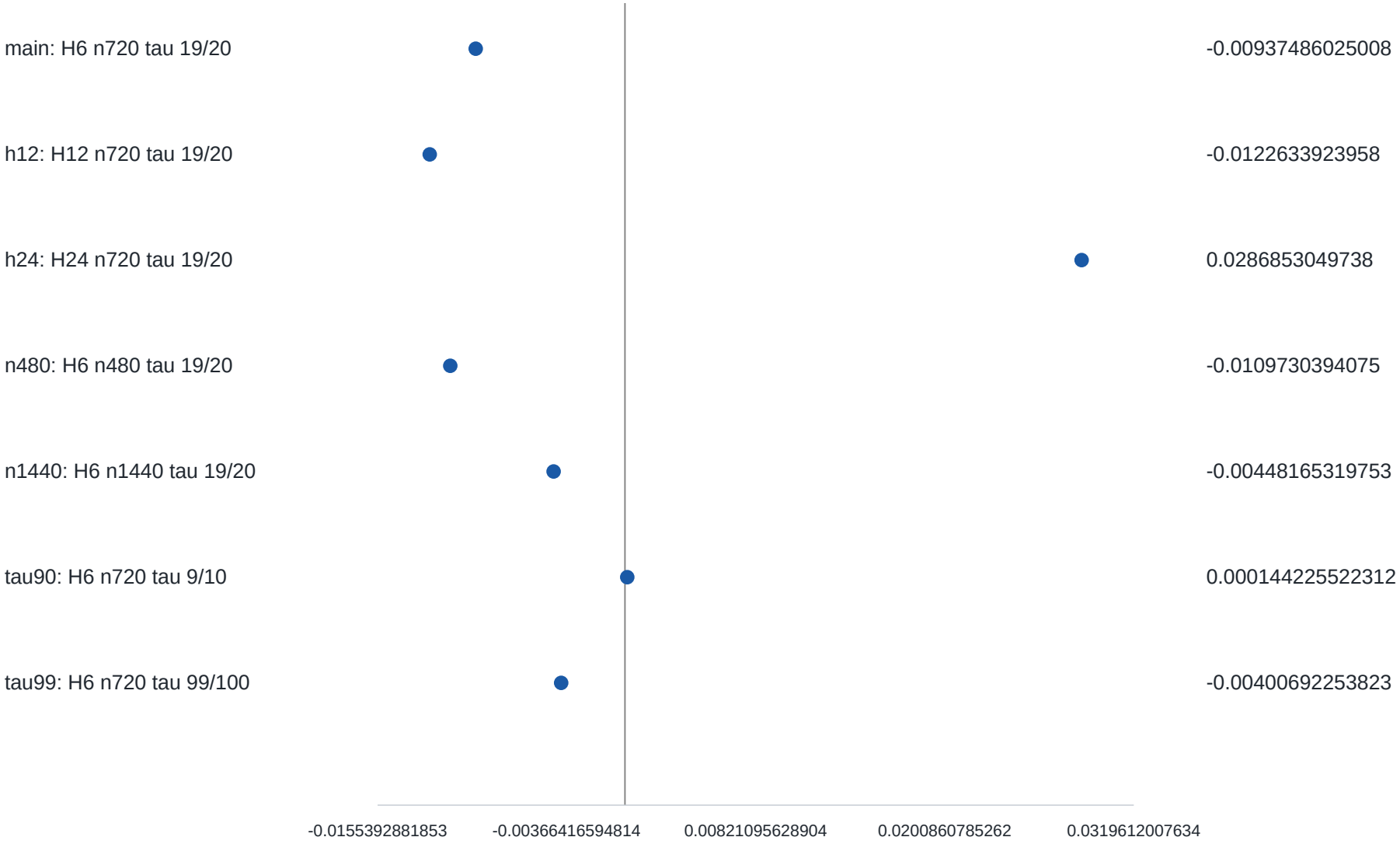
Figure 1. Original seven-cell F-minus-S paired means. Negative values favor F for that fixed preference. Different quantile preferences are not pooled into an overall winner.

Figure 2. All 12 fixed exploratory nearest-minus-recent loss contrasts. Negative values favor nearest retrieval at the same horizon, count and representation. The marked primary is H6, m360, K; none of these contrasts establishes superiority over the original full-window leaders.

Figure 3. Every monthly primary K retrieval contrast at both horizons, including adverse January 2016. Counts are source-record origins. Calendar groups are descriptive, not independent validations or switching regimes.

Within-cell paired mean loss differences

F minus S; USD millions. Negative values favor F within the same fixed cell.



Seven declared cells in fixed order; no comparison of raw losses across different tau objectives.

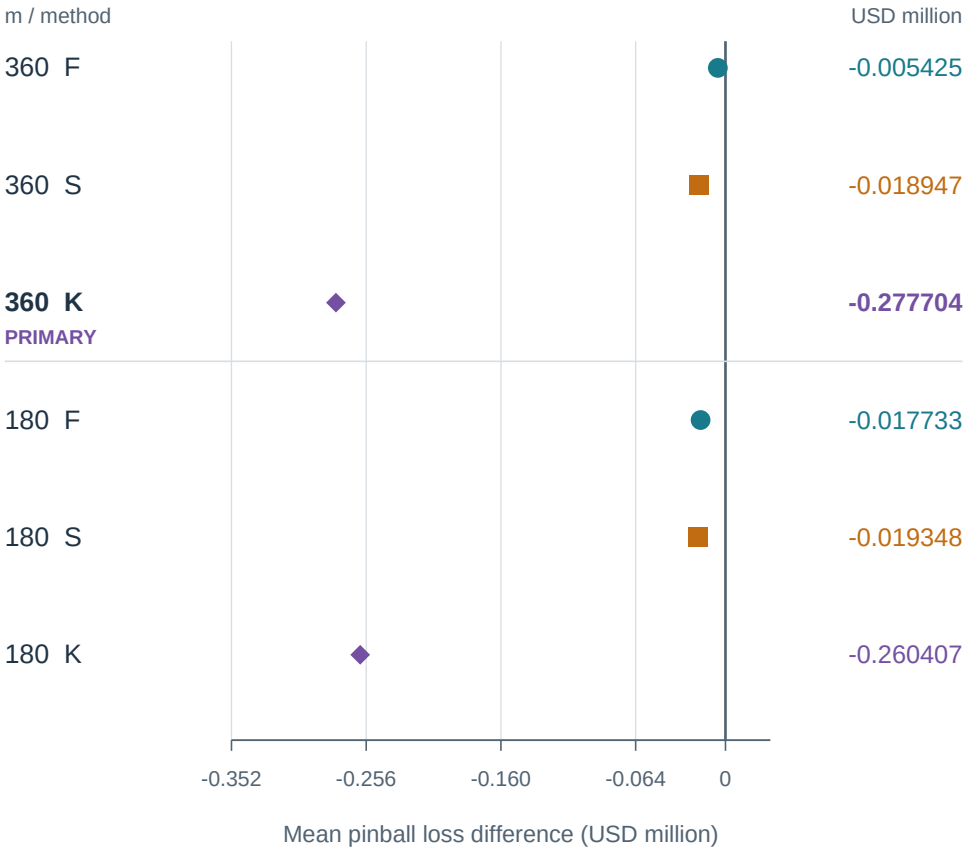
Points are finite-record paired means; the zero line is descriptive, not an inferential threshold.

Nearest versus recent: all frozen loss contrasts

Exploratory paired means | nearest minus recent | negative favors nearest

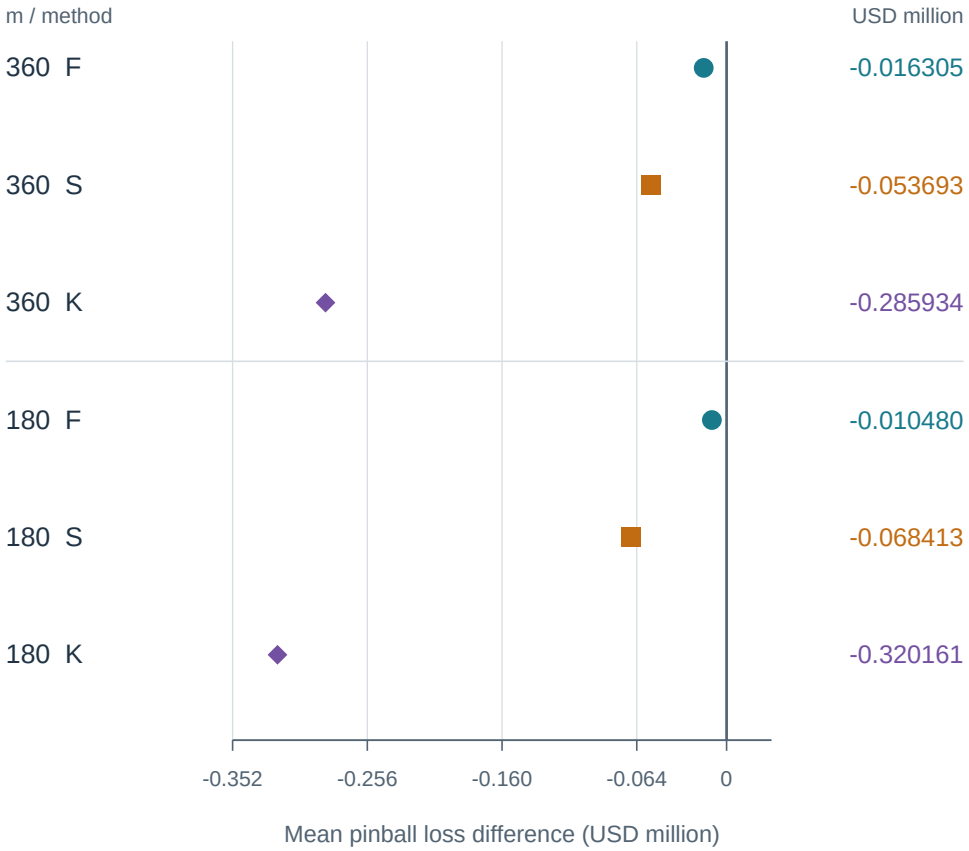
H = 6 records

6,625 paired origins in every row



H = 24 records

6,625 paired origins in every row



F S K

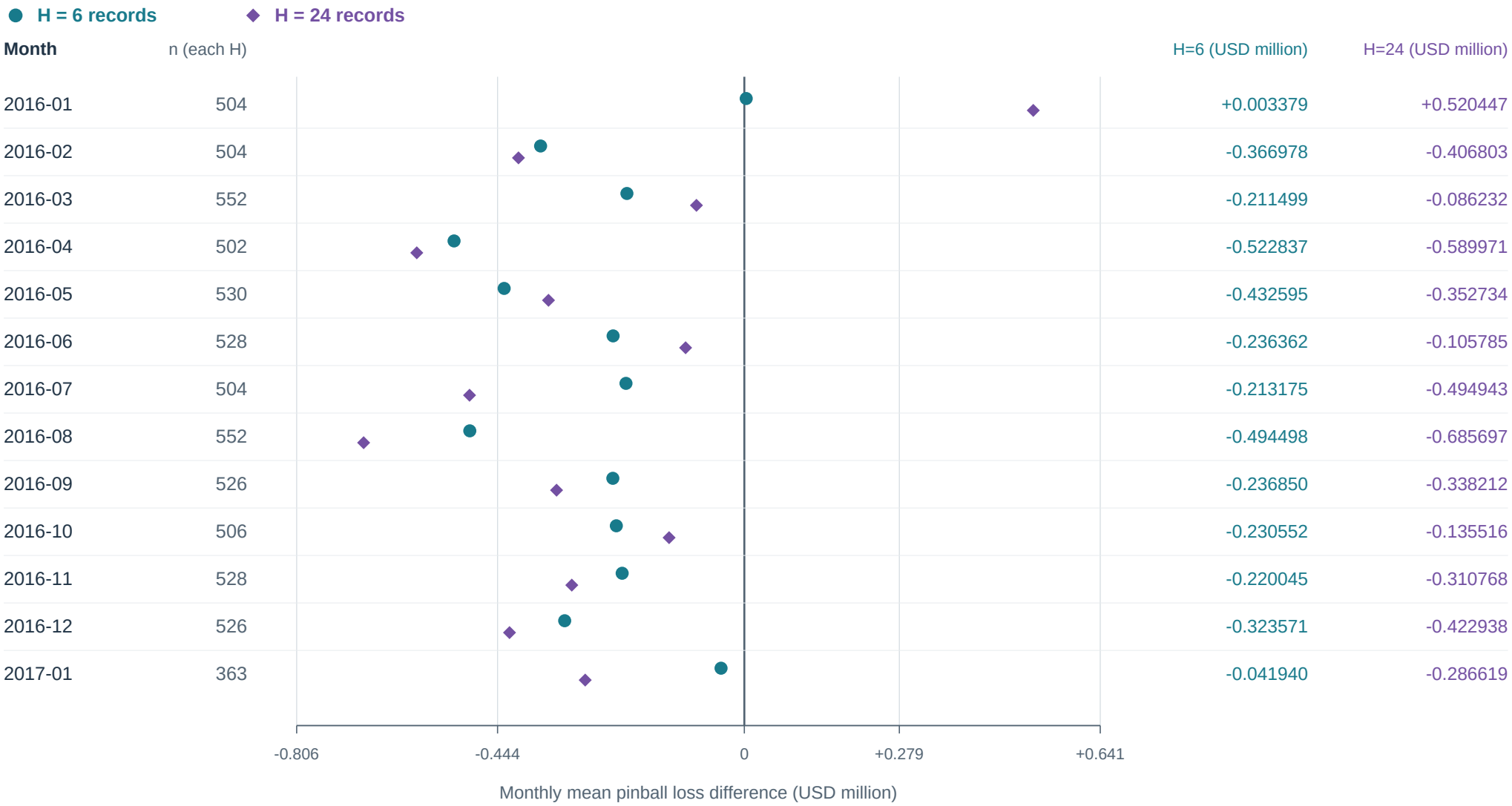
PRIMARY marks the prespecified H=6, m=360, K contrast; it does not imply an overall ranking.

All 12 frozen nearest-minus-recent aggregate means are shown. No uncertainty intervals are displayed.

Source: accepted public comparisons.csv | exploratory-state-selection-20261021

Monthly primary contrasts: descriptive / exploratory

Exploratory paired means | nearest360 K minus recent360 K | negative favors nearest



All 13 calendar months are retained, including positive January 2016 at both horizons.

Descriptive summaries of the inspected evaluation record. Zero marks equal average pinball loss.

Source: accepted public monthly.csv | primary nearest360 K minus recent360 K